

Supplementary Materials for ”Evidence Decision Loops”

Updated: 2026-07-16

This supplement supports the English JBI manuscript. It contains dataset composition, trace completeness, full metric matrices, bootstrap confidence intervals, schema and prompt indexes, supplementary model and ablation results, blinded clinical-expert audit design and results, baseline-fairness controls, and reproducibility paths.

Supplementary Table S1. Source composition of candidate, main evaluation, and non-retained records

Split	Records	SciFact	PubMedQA	NLI4CT	Required evidence segments
Candidate set	460	150	150	160	1635
Main evaluation set	390	135	125	130	1413
Non-retained or further-adjudication records	70	15	25	30	222

Supplementary Table S2. Source-label distribution

Source	Label	Candidate set	Main evaluation set
SciFact	SUPPORT	50	42
SciFact	CONTRADICTION	50	44
SciFact	NEI	50	49
PubMedQA	yes	50	43
PubMedQA	no	50	43
PubMedQA	maybe	50	39
NLI4CT	Entailment	80	65
NLI4CT	Contradiction	80	65

Supplementary Table S3. Main evaluation trace completeness

Model	Setting	Expected traces	Observed traces	Methods	Invalid	Missing	Provider failures	Parse or fallback flags
gpt-5.5	Given evidence	1560	1560	direct,cot,react,edl	0	0	0	none
gpt-5.5	Evidence seeking	780	780	react,edl	0	0	0	forced_submit:83

Supplementary Table S4. Evaluation metric dictionary

Metric	Scope and role	Definition
Decision accuracy	All methods; final-label correctness	Exact agreement between the submitted decision and the source-native gold decision.
Primary metric	All methods; joint answer-and-evidence success	Binary record-level score equal to 1 only when the final decision is correct, citations are valid, and required evidence recall is 1.
Required evidence recall	All methods; evidence coverage	Proportion of gold required evidence segments covered by valid final citations.
Required evidence precision	All methods; citation specificity	Proportion of valid final citations that correspond to required evidence segments.
Citation validity	All methods; structural citation correctness	Whether every cited evidence segment exists in the current record and contains valid <code>documentId</code> and <code>segmentId</code> fields.
Search-output coverage	Evidence-seeking setting; diagnostic process metric	Proportion of required evidence segments returned by search outputs before reading. This metric is used for failure localization rather than as a primary EDL-ReAct performance endpoint.
Read recall	Evidence-seeking setting; inspected-evidence coverage	Proportion of required evidence segments opened by read tools or EDL inspection.
Process validity	EDL and process traces; audit conformance	Whether externally recorded process steps satisfy the evaluation protocol; hidden model reasoning is not scored.
Human review rate and reasons	EDL review routing; review-queue characterization	Proportion of records flagged for human review and the associated review reasons.
Invalid decision rate	All methods; output-contract failure	Proportion of submitted decisions outside the source-native allowed label space or not parseable as a final decision.

Supplementary Table S5. EDL review routing in the evidence-seeking setting

Source	Traces	Review rate	Insufficient evidence	Forced source label	Material gap or conflict
All	390	0.285	101	49	23
SciFact	135	0.385	52	0	0
PubMedQA	125	0.160	20	20	0
NLI4CT	130	0.300	29	29	23

Supplementary Table S6. SciFact evidence-seeking cross-model replication

Model	Method	Traces	Accuracy	Primary metric	Required recall	Search-output coverage	Read recall	Parse or fallback flags
gpt-5.5	ReAct	135	0.993	0.859	0.874	0.000	0.985	forced_submit:4
gpt-5.5	EDL	135	0.948	0.874	0.933	0.967	0.967	none
MiniMax-M3	ReAct	135	0.904	0.800	0.900	0.937	0.941	forced_submit:31
MiniMax-M3	EDL	135	0.889	0.830	0.948	0.974	0.974	none

Supplementary Table S7. Given-evidence results by required evidence segment count

Method	Required evidence segments	Traces	Accuracy	Primary metric	Required evidence recall
Direct	1	250	0.856	0.788	0.932
Direct	2	42	0.976	0.524	0.762
Direct	3--5	54	0.981	0.241	0.634
Direct	6+	44	0.955	0.000	0.305
CoT	1	250	0.884	0.820	0.936
CoT	2	42	0.976	0.476	0.738
CoT	3--5	54	0.981	0.259	0.628
CoT	6+	44	0.932	0.000	0.371
ReAct	1	250	0.888	0.828	0.940
ReAct	2	42	0.976	0.619	0.798
ReAct	3--5	54	0.981	0.407	0.715
ReAct	6+	44	0.932	0.000	0.348
EDL	1	250	0.852	0.816	0.964
EDL	2	42	1.000	0.952	0.964
EDL	3--5	54	0.981	0.889	0.941
EDL	6+	44	0.955	0.318	0.735

Supplementary Table S8. Evidence-seeking results by required evidence segment count

Method	Required evidence segments	Traces	Accuracy	Primary metric	Required evidence recall	Search-output coverage	Read recall
ReAct	1	250	0.952	0.892	0.940	0.000	0.992
ReAct	2	42	0.952	0.571	0.690	0.095	0.821
ReAct	3--5	54	0.870	0.148	0.456	0.228	0.668
ReAct	6+	44	0.841	0.000	0.290	0.160	0.702
EDL	1	250	0.856	0.828	0.972	0.984	0.984
EDL	2	42	0.952	0.833	0.917	0.964	0.964
EDL	3--5	54	0.944	0.630	0.848	0.884	0.884
EDL	6+	44	0.886	0.182	0.535	0.705	0.705

Supplementary Table S9. Schema and field-definition index

Artifact	Path	Main role
EDL decision record schema	src/schema/decision-record.schema.json	Defines benchmark record fields and prediction fields, including final decision, cited claims, cited evidence, claim evaluations, gaps, conflicts, and expert review.
Live comparison trace schema	src/schema/live-comparison-trace.schema.json	Defines trace-level fields including <code>recordId</code> , <code>method</code> , <code>model</code> , <code>reasoningEffort</code> , <code>promptSha256</code> , <code>steps</code> , <code>finalDecision</code> , <code>citedEvidenceIds</code> , <code>requiredHumanReview</code> , <code>edlGuard</code> , <code>processCheck</code> , and <code>scores</code> .
V4 audit output schema	src/schema/react-clinical-source-v4-audit-output.schema.json	Constrains pre-evaluation consistency-audit outputs, including audit decision, suggested decision, confidence, rationale, and segment-level cited evidence.
Human audit study schema	results/human-audit-sciFact-nli4ct-balanced-docker/study.json	Records the completed blind-audit sample size, source targets, rater count, seed, source data, and trace path.

Supplementary Table S10. Prompt and template index

Method or stage	Path	Contract summary
V4 base evidence contract	src/prompts/v4_evidence_verification_base.md	Use only frozen evidence; do not use external knowledge; final decision must be one of <code>allowedDecisions</code> ; citations must include <code>documentId</code> and <code>segmentId</code> ; return JSON only.
Direct	src/prompts/v4_method_direct.md	Read the record prompt and submit one JSON answer.
CoT	src/prompts/v4_method_cot.md	Use internal reasoning if needed, but return only final JSON.
ReAct	src/prompts/v4_method_react.md	Search before reading, continue when evidence is insufficient, and submit a JSON-compatible decision with segment-level citations.
EDL	src/prompts/v4_method_edl.md	Search, read, assess sufficiency, validate citations, repair invalid citations, and submit one final decision.
LLM-assisted pre-evaluation consistency audit	src/scripts/run_v4_llm_audit.js	Judges whether source-native gold decisions are supported by frozen segment-level evidence and emits structured JSON.

Supplementary Table S11. Pre-evaluation consistency-audit filtering summary

Source	Candidate records	Main inclusion rule	Included	Non-retained or further adjudication	Audit artifact
SciFact	150	<code>auditDecision=agree</code>	135	15	results/v4-llm-audit-full/v4-llm-audit-summary.md
PubMedQA	150	<code>auditDecision=agree</code>	125	25	results/v4-llm-audit-full/v4-llm-audit-summary.md
NLI4CT	160	<code>includeInMain=true</code>	130	30	results/v4-llm-audit-nli4ct-fultrial-rawline-validation-160-full/v4-llm-audit-summary.md
All	460	Unified pre-evaluation consistency audit and canonical dataset construction	390	70	results/final-canonical-audit-merge/final-relevant-audit-summary.md

The final-relevant audit merge is provided in results/final-canonical-audit-merge/. It combines the SciFact/PubMedQA audit records from results/v4-llm-audit-full/v4-llm-audit-results.jsonl with the corrected NLI4CT audit records from results/v4-llm-audit-nli4ct-fultrial-rawline-validation-160-full/v4-llm-audit-results.jsonl. The NLI4CT canonical audit over 160 records produced 138 agree, 11 disagree, and 11 uncertain audit decisions; 130 records were marked `includeInMain=true`. The audit recorded 24 partial-evidence cases, 16 granularity issues, and no material conflicts.

Supplementary Table S12. Derived exclusion-reason taxonomy for non-retained records

The 70 non-retained records were assigned to a derived, mutually exclusive taxonomy from the final-relevant audit outputs. Categories were applied in priority order: label-evidence disagreement, unclear label support, insufficient or partial required evidence, evidence granularity or missing-context problems, and other scoring-protocol refinement. The taxonomy summarizes why records were not included in the programmatically scored main evaluation set; it was not used as a model-performance endpoint.

		Exclusion reason	Records
		Label-evidence disagreement	50
		Unclear label support	12
		Insufficient or partial required evidence	8
		Evidence granularity or missing-context problem	0
		Other scoring-protocol refinement	0

Source	Label-evidence disagreement	Unclear label support	Insufficient or partial required evidence	Total non-retained
SciFact	14	1	0	15
PubMedQA	25	0	0	25
NLI4CT	11	11	8	30
All	50	12	8	70

Supplementary Appendix A. Cohort construction and execution chronology

The comparative cohort was defined before comparative execution from source-record audit metadata, without access to Direct, CoT, ReAct, or EDL outputs. The 390 audit-retained records were then used as the common input cohort for every workflow. The 70 records not retained by the audit were not rerun or rescored post hoc. This appendix reports the traceability artifacts supporting that chronology; it is not a prospective preregistration.

Stage	Traceability artifact	Role
Candidate-record audit	results/final-canonical-audit-merge/final-relevant-audit-summary.md and its linked record-level audit outputs	Documents audit decisions across the 460 candidate records and the source-record basis for retention.
Retained cohort	src/data/react-clinical-source-v4-main-three-source-canonical-nli4ct.jsonl	Canonical 390-record audit-retained comparative cohort.
Cohort manifest	results/react-clinical-source-v4-main-three-source-canonical-nli4ct-manifest.json	Records the main dataset composition and the paths to comparison outputs.
Given-evidence comparison	results/live-comparison-v4-main-three-source-canonical-nli4ct-frozen-evidence-full/live-comparison-v4-results.json	Records the 390-record input cohort across Direct, CoT, ReAct, and EDL.
Evidence-seeking comparison	results/live-comparison-v4-main-three-source-canonical-nli4ct-evidence-seeking-full/live-comparison-v4-results.json	Records the same 390-record input cohort across ReAct and EDL.
Versioned public release	Evidence Decision Loops release v0.2.8, including SHA256SUMS-v0.2.8.txt	Fixed reproducibility target for the audit, cohort, manifests, comparison outputs, and final manuscript assets; it is not the mutable repository head.

Supplementary Table S13. Full given-evidence source x method x metric matrix

Source	Method	Traces	Accuracy	Primary metric	Required recall	Required precision	Citation validity	Review rate
SciFact	Direct	135	0.948	0.800	0.859	0.645	0.985	0.015
SciFact	CoT	135	0.963	0.800	0.856	0.649	0.993	0.007
SciFact	ReAct	135	0.956	0.815	0.867	0.648	1.000	0.000
SciFact	EDL	135	0.963	0.881	0.922	0.621	1.000	0.415
PubMedQA	Direct	125	0.760	0.760	1.000	1.000	1.000	0.000
PubMedQA	CoT	125	0.800	0.800	1.000	1.000	1.000	0.000
PubMedQA	ReAct	125	0.816	0.816	1.000	1.000	1.000	0.000
PubMedQA	EDL	125	0.744	0.744	1.000	1.000	1.000	0.160
NLI4CT	Direct	130	0.977	0.223	0.551	0.917	1.000	0.000
NLI4CT	CoT	130	0.969	0.238	0.575	0.919	1.000	0.000
NLI4CT	ReAct	130	0.969	0.331	0.619	0.916	1.000	0.000
NLI4CT	EDL	130	0.977	0.723	0.886	0.786	1.000	0.154

Supplementary Table S14. Full evidence-seeking source x method x metric matrix

Source	Method	Traces	Accuracy	Primary metric	Final required recall	Required precision	Search-output coverage	Read recall	Citation validity	Review rate
SciFact	ReAct	135	0.993	0.859	0.874	0.643	0.000	0.985	1.000	0.000
SciFact	EDL	135	0.948	0.874	0.933	0.608	0.967	0.967	1.000	0.385
PubMedQA	ReAct	125	0.912	0.912	1.000	1.000	0.000	1.000	1.000	0.000
PubMedQA	EDL	125	0.768	0.768	1.000	1.000	1.000	1.000	1.000	0.160
NLI4CT	ReAct	130	0.877	0.192	0.449	0.727	0.180	0.703	0.992	0.008
NLI4CT	EDL	130	0.923	0.538	0.768	0.864	0.844	0.844	1.000	0.300

Supplementary Table S15. Sensitivity endpoints for the evidence-seeking EDL-ReAct comparison

These endpoints separate final-label accuracy from evidence-process and answer-evidence binding measures. Source-balanced primary metric is the unweighted mean of the SciFact, PubMedQA, and NLI4CT source-level primary metrics reported in Supplementary Table S14. Differences are EDL minus ReAct.

Endpoint	What it tests	ReAct	EDL	Difference
Final-label accuracy	Answer correctness only	0.928	0.882	-0.046
Primary metric	Correct answer plus complete valid required-evidence binding	0.654	0.728	0.074
Required evidence recall	Coverage of required evidence in final citations	0.773	0.900	0.127
Required evidence precision	Proportion of final citations that are required evidence	0.786	0.819	0.033
Citation validity	Structural validity of cited document and segment identifiers	0.997	1.000	0.003
Read recall	Required evidence opened through read/inspection steps	0.896	0.937	0.041
Source-balanced primary metric	Whether primary-metric direction depends on source mix	0.654	0.727	0.072

Supplementary Table S16. PubMedQA evidence-seeking label-calibration profile

All PubMedQA evidence-seeking traces had complete required evidence recall, complete read recall, and valid final citations. This table therefore isolates final-label calibration errors under complete evidence availability. Rows are gold source-native labels and columns are submitted labels.

ReAct confusion matrix:

Gold label	Submitted yes	Submitted no	Submitted maybe	Accuracy
yes	43	0	0	1.000
no	0	43	0	1.000
maybe	10	1	28	0.718

EDL confusion matrix:

Gold label	Submitted yes	Submitted no	Submitted maybe	Accuracy
yes	39	0	4	0.907
no	0	42	1	0.977
maybe	18	6	15	0.385

Error-overlap and routing summary:

Error profile	Value
ReAct wrong decisions	11
EDL wrong decisions	29
Wrong for both methods	9
Wrong for EDL only	20
Wrong for ReAct only	2
EDL errors on gold maybe records submitted as yes/no	24
EDL errors on gold yes/no records submitted as maybe	5
EDL PubMedQA review reasons: insufficient_evidence	20
EDL PubMedQA review reasons: forced_source_label_without_abstain	20

The reproducible artifact for this table is generated by `node src/scripts/analyze_v4_pubmedqa_calibration.js --write` and stored as `results/v4-pubmedqa-calibration-profile.json` and `results/v4-pubmedqa-calibration-profile.md`.

Supplementary Table S17. MiniMax-M3 three-source supplementary evaluation

Setting	Method	Traces	Accuracy	Primary metric	Required recall	Required precision	Search-output coverage	Read recall	Citation validity	Review rate	Invalid decision
Given evidence	Direct	390	0.877	0.562	0.800	0.828	n/a	n/a	0.995	0.005	0.000
Given evidence	CoT	390	0.859	0.546	0.797	0.834	n/a	n/a	0.995	0.005	0.000
Given evidence	ReAct	390	0.864	0.562	0.816	0.819	n/a	n/a	0.997	0.003	0.000
Given evidence	EDL	390	0.874	0.718	0.906	0.785	n/a	n/a	0.992	0.469	0.000
Evidence seeking	ReAct	390	0.887	0.615	0.814	0.796	0.909	0.917	0.997	0.003	0.000
Evidence seeking	EDL	390	0.854	0.685	0.890	0.795	0.932	0.932	0.997	0.526	0.003

MiniMax-M3 is a supplementary evaluation and not a co-primary replacement for the **gpt-5.5** main evaluation.

Supplementary Table S18. EDL component ablation matrix

The Full EDL reference for each source was reused from the corresponding main evidence-seeking experiment; no additional model inference was performed for either Full reference row. The three SciFact feature-removal variants were newly executed on the same 135 records using `gpt-5.5`, medium reasoning effort, BM25, controller-fixed top-5 retrieval, a maximum of three evidence rounds, a minimum of two rounds and three viewed segments before declaring sufficiency, and no pre-submission citation self-check. Each variant was configured to remove the named controller component within its source-specific runtime configuration. The NLI4CT variants retained source-aware query planning, BM25 neighbor expansion, and controller-fixed top-5 retrieval. The optional citation self-check was retained except in the NLI4CT no-citation-validation variant. In the no-citation-validation variants, the cited-identifier output rule and citation-error validation/repair gate were disabled, while post-submission benchmark scoring still calculated citation-identifier validity. The comparisons are therefore descriptive within-source configuration contrasts, not a pooled ablation under one uniform cross-source runtime configuration or estimates of isolated component effects. Each record-variant cell contains one completed output; repeated-run variability was not estimated.

Source	Variant	Traces	Accuracy	Primary metric	Required recall	Required precision	Citation validity	Review rate	Main review reasons
SciFact	Full EDL reference	135	0.948	0.874	0.933	0.608	1.000	0.385	insufficient_evidence:52
SciFact	Without iterative seeking	135	0.933	0.785	0.822	0.602	1.000	0.481	insufficient_evidence:65, material_gap:50
SciFact	Without sufficiency assessment	135	0.948	0.815	0.870	0.637	0.985	0.015	invalid_citation:2
SciFact	Without citation validation	135	0.941	0.059	0.067	0.731	0.067	0.948	insufficient_evidence:55, invalid_citation:126, material_conflict:1, material_gap:42, missing_citation:9
NLI4CT	Full EDL reference	130	0.923	0.538	0.768	0.864	1.000	0.300	forced_source_label_without_abstain:29, insufficient_evidence:29, material_conflict:13, material_gap:10
NLI4CT	Without iterative seeking	130	0.869	0.423	0.691	0.852	1.000	0.385	forced_source_label_without_abstain:44, insufficient_evidence:44, material_conflict:9, material_gap:23
NLI4CT	Without sufficiency assessment	130	0.931	0.577	0.777	0.877	1.000	0.000	none
NLI4CT	Without citation validation	130	0.923	0.115	0.144	0.801	0.200	0.831	forced_source_label_without_abstain:34, insufficient_evidence:34, invalid_citation:104, material_conflict:13, material_gap:14

Supplementary Table S19. Paired bootstrap confidence intervals for key evidence-seeking metrics

Intervals were computed from existing record-level trace scores with 10,000 record-level bootstrap resamples. No additional model inference was performed. Differences are EDL minus ReAct.

Metric	ReAct	EDL	Difference	95% CI
Primary metric	0.654	0.728	0.074	[0.026, 0.123]
Required evidence recall	0.773	0.900	0.127	[0.096, 0.158]
Required evidence precision	0.786	0.819	0.033	[0.006, 0.061]
Search-output coverage (diagnostic)	0.060	0.937	0.877	[0.845, 0.907]
Read recall	0.896	0.937	0.041	[0.015, 0.068]
Decision accuracy	0.928	0.882	-0.046	[-0.079, -0.013]

The full artifact is results/v4-main-bootstrap-ci.json, with a rendered table in results/v4-main-bootstrap-ci.md.

Supplementary Table S20. Blinded clinical-expert audit design and results

Item	Current value
Main audit bundle	results/human-audit-sciifact-nli4ct-balanced-docker/
Created at	2026-07-02T17:59:02.015Z
Random seed	eviagent-human-audit-sciifact-nli4ct-balanced-v1
Sample size	60 cases
Source stratification	30 SciFact cases and 30 NLI4CT cases
Sampling design	Selected cases balanced by preclassified output-success and output-failure patterns; see <code>study.json</code> for source-specific targets
Clinical experts	8 raters (RATER_A--RATER_H)
Assignment mode	all_raters_balanced
Raters per case/format	4
Method anonymization	Anonymous report formats; method names and benchmark-derived correctness fields hidden during rating
Rater instructions	human-audit-expert-instructions.md; defines acceptance/interception and the 1--5 confidence and cognitive-load anchors
Input data	src/data/react-clinical-source-v4-main-three-source-canonical-nli4ct.jsonl
Input traces	results/live-comparison-v4-main-three-source-canonical-nli4ct-evidence-seeking-full/live-comparison-v4-traces.jsonl
Raw submissions	485 rows, including 5 superseded revisions
Final analytic ratings	480 ratings: 240 EDL and 240 ReAct
Revision rule	Retain the highest revision number for each assignment
Missing data and review time	No final analytic record was missing an outcome, rating, or elapsed-time field; no duration-based exclusions were applied (range, 16.2--356.9 s)
Failure-report ratings per method	120 scoring-defined decision-or-evidence-binding failures
Acceptable-report ratings per method	120 reports satisfying both correct decision and primary metric
Derived descriptive results	exports/human-audit-results.json and exports/human-audit-results.md
Attempted crossed-model analysis	src/scripts/analyze_human_audit_mixed_effects.py; the two fits did not converge and are not reported inferentially
Reporting status	Completed clinical-expert audit; descriptive results are reported in the main manuscript; attempted clustered sensitivity models are documented in Supplementary Note S21 as non-convergent

The raw response file and rater credentials remain in the access-controlled `private/` directory and are not part of the public submission source package. An older SciFact-only pilot bundle exists in the working project but is not included in this release and is not used as the balanced clinical-expert audit result.

Supplementary Note S21. Attempted crossed random-effects analysis of the blinded clinical-expert audit

We attempted separate logistic random-intercept models for the two binary outcomes: `interception ~ method + (1 | rater) + (1 | case)`. Both fits were marked non-convergent in the frozen analysis artifact. We therefore do not report model-based odds ratios, confidence intervals, or p values. The rating-level summaries below are descriptive only and do not estimate real-world error prevalence, clinical safety, prospective reviewer effectiveness, or deployment performance.

Outcome subset		EDL	ReAct	Attempted model status
Interception among preclassified benchmark-failure reports	73/120 (60.8%)		40/120 (33.3%)	Non-convergent; no inferential estimate reported
False interception among preclassified acceptable reports	2/120 (1.7%)		12/120 (10.0%)	Non-convergent; no inferential estimate reported

The frozen analysis artifacts remain available for transparency, but are not used as inferential evidence in the manuscript.

Supplementary Table S22. Trace-derived qualitative example

This illustrative example is taken from the frozen evidence-seeking traces and is not used as a separate outcome analysis. It shows how the primary metric distinguishes final-label correctness from complete evidence binding.

Field	ReAct	EDL
Record	scifact-v4-0014	scifact-v4-0014
Source task	SciFact statement verification: a diminished ovarian reserve does not solely indicate infertility in an a priori non-infertile population	Same record
Gold decision	SUPPORT	SUPPORT
Submitted decision	SUPPORT	SUPPORT
Decision correct	1	1
Required evidence ID	scifact-doc-13497630:seg-8	scifact-doc-13497630:seg-8
Final cited evidence IDs	scifact-doc-13497630:seg-2; scifact-doc-13497630:seg-11	scifact-doc-13497630:seg-11; scifact-doc-13497630:seg-2; scifact-doc-13497630:seg-8; scifact-doc-13497630:seg-9; scifact-doc-13497630:seg-12
Required evidence recall	0.000	1.000
Required evidence precision	0.000	0.200
Citation validity	1	1
Primary metric	0	1
Interpretation	The final label was correct, but the submitted citations did not include the required evidence segment.	The final label was correct and the final citations included the required evidence segment, although with lower citation precision because additional segments were cited.

The required segment reports that elevated serum FSH, a marker of diminished ovarian reserve, was not associated with a significantly different predicted probability of conception over 6 or 12 cycles. The example illustrates why final-label accuracy alone is insufficient for evaluating evidence-supported decision records.

Supplementary Table S23. Reproducibility configuration and baseline-fairness controls

This table summarizes the configuration needed to interpret the EDL-ReAct comparison as a workflow-level comparison with matched evidence access and scoring. It is not a claim that ReAct and EDL used identical controller complexity, reasoning-effort configuration, or computational resources.

Dimension	Main configuration
Main dataset	src/data/react-clinical-source-v4-main-three-source-canonical-nli4ct.jsonl; 390 audit-retained records from SciFact, PubMedQA, and NLI4CT
Dataset manifest	results/react-clinical-source-v4-main-three-source-canonical-nli4ct-manifest.json
Main model backend	gpt-5.5 for the primary automated evaluation
Trace metadata	Trace files retain model , reasoningEffort , promptSha256 , method, record ID, submitted decision, cited evidence IDs, process checks, EDL guard fields, and score fields
Given-evidence methods	Direct, CoT, ReAct, and EDL; 390 records x 4 methods = 1560 traces
Evidence-seeking methods	ReAct and EDL; 390 records x 2 methods = 780 traces
Matched comparison fields	Same records, source-native label spaces, admissible citation identifiers, fixed evidence libraries, output schema, and scoring code within each setting
Evidence access	Record-scoped fixed evidence libraries only; no open-web retrieval
Search and read tools	search_segments returns candidate segment identifiers/snippets from the fixed record library; read_segment opens frozen segment text before citation
Retrieval configuration	BM25 search; short abstract sources used default top-5 search, while longer NLI4CT trial reports used neighbor expansion and section-aware logic
ReAct baseline role	Strong tool-using / agentic RAG baseline, not an answer-only strawman
ReAct fallback handling	Forced-submit fallback when the configured tool budget was exhausted; 83 evidence-seeking ReAct traces used this fallback and remained in scoring
EDL controller additions	Evidence-sufficiency assessment, sufficiency-driven query updating, citation validation, process validation, gap/conflict recording, and derived human-review routing
Compute interpretation	Matched records, evidence access, and scoring only; not matched for reasoning-effort configuration, total model calls, tool calls, latency, token use, monetary cost, or controller complexity
Prompt artifacts	src/prompts/v4_method_direct.md, src/prompts/v4_method_cot.md, src/prompts/v4_method_react.md, src/prompts/v4_method_edl.md, and src/prompts/v4_evidence_verification_base.md
Scoring contract	Source-native final-label accuracy, required evidence recall/precision, citation validity, process validity, read recall in evidence seeking, and the joint answer-and-evidence primary metric
Statistical inference	Record-level paired nonparametric bootstrap with 10,000 resamples, computed from existing trace scores without additional model inference
Supplementary checks	MiniMax-M3 cross-model evaluation and SciFact/NLI4CT EDL component ablations are supplementary analyses, not replacements for the gpt-5.5 main evaluation

Supplementary Table S24. Reproducibility paths and verification entry points

Category	Path or command	Role
Main dataset manifest	results/react-clinical-source-v4-main-three-source-canonical-nli4ct-manifest.json	Records main dataset composition, labels, required evidence distribution, and result paths.
Main dataset	src/data/react-clinical-source-v4-main-three-source-canonical-nli4ct.jsonl	Canonical three-source main dataset.
gpt-5.5 given-evidence results	results/live-comparison-v4-main-three-source-canonical-nli4ct-frozen-evidence-full/live-comparison-v4-results.json	390 records x 4 methods = 1560 traces.
gpt-5.5 evidence-seeking results	results/live-comparison-v4-main-three-source-canonical-nli4ct-evidence-seeking-full/live-comparison-v4-results.json	390 records x 2 methods = 780 traces.
Dual-setting main summary	results/live-comparison-v4-main-three-source-canonical-nli4ct-dual-setting-summary.md	Source for main given-evidence and evidence-seeking summaries.
MiniMax-M3 summary	results/live-comparison-v4-minimax-m3-canonical-nli4ct-dual-setting-summary.md	Three-source supplementary model evaluation.
EDL ablation summary	results/edl-ablation-s18-final-summary.md	SciFact and NLI4CT component ablation.
Bootstrap CI script	node src/scripts/analyze_v4_bootstrap_ci.js --write --iterations 10000	Generates record-level bootstrap confidence intervals from existing trace scores without model inference.
Bootstrap CI artifact	results/v4-main-bootstrap-ci.json and results/v4-main-bootstrap-ci.md	Main paired confidence intervals for evidence-seeking EDL-ReAct comparisons and supplementary given-evidence comparisons.
PubMedQA calibration script	node src/scripts/analyze_v4_pubmedqa_calibration.js --write	Regenerates the PubMedQA confusion matrices and label-calibration profile from the canonical dataset and evidence-seeking traces.
PubMedQA calibration artifact	results/v4-pubmedqa-calibration-profile.json and results/v4-pubmedqa-calibration-profile.md	Source for Supplementary Table S16.
JB1 V4 reproducibility check	npm run paper:jb1-v4:reproducibility:check	Verifies the current JBI/V4 artifact paths, core record/trace counts, and PubMedQA calibration artifact drift.
Human audit build	npm run human-audit:build	Generates balanced blind-audit study bundle.
Human audit analysis	node src/scripts/analyze_human_audit_study.js --study-dir results/human-audit-scifact-nli4ct-balanced-docker	Regenerates the locked derived audit summaries from final ratings.
Repository tests	npm test	Project-level test entry point.
English manuscript compile	cd content && pdflatex -interaction=nonstopmode -halt-on-error EDL_JBI_manuscript_en.tex	Compiles the English manuscript.

Trace metadata note: main trace files retain **model** and **reasoningEffort** fields. The manuscript therefore describes the main analysis as the **gpt-5.5** evaluation and reports exact run metadata through the reproducibility package rather than asserting a single uniform reasoning-effort value for every merged trace.

Supplementary Note S25. Release and controlled-access statement

The fixed reproducibility target for this manuscript is the public Evidence Decision Loops v0.2.8 release, which provides the source package, final manuscript assets, and **SHA256SUMS-v0.2.8.txt** checksum manifest. The tagged repository contains the benchmark, frozen traces, result summaries, code, prompts, schemas, and permitted expert-audit materials.

Completed expert-response records, rater credentials, access links, and direct identifiers are not publicly distributed. Requests for deidentified rating data and corresponding audit-analysis outputs may be directed to the corresponding author through the submission contact. Requests are considered under the ethics determination (ZS09437881), informed-consent conditions, and institutional data-governance requirements; this release does not authorize unrestricted redistribution of those records.